Lab 8.3 – Static Route Troubleshooting (Packet Tracer)

This activity comes with an accompanying Packet Tracer file with a partially configured network. Download the Packet Tracer file from the Animospace assignment page.

Objectives

Part 1: Evaluate Network Operation

Part 2: Gather Information, Create an Action Plan, and Implement Corrections

Background / Scenario

All the network devices in this activity have been preconfigured to include intentional errors that are preventing the network from routing as intended. Your task is to evaluate the network, identify, and correct the configuration errors to restore full connectivity. You may find errors with the route statements or with other configurations that impact the accuracy of the route statements.

Note: The static routing approach used in this activity is used to assess your ability to configure different types of static routes only. This approach may not reflect networking best practices.

Instructions

Part 1: Part 1: Evaluate Network Operation

Use the ping and/or traceroute commands from the router to test the following criteria and record the results.

- Traffic from R1 to the 172.16.2.1 address on R2 uses the next hop 192.168.0.14.
- Traffic from R1 to the 209.165.200.129 address on R2 uses the next hop 192.168.0.30.
- When the G0/0/0 interface on R1 is shut down, traffic from R1 to the 172.16.2.1 address on R2 uses the next hop 192.168.0.30.
- Traffic from R2 to the 2001:db8:acad:171::1 address on R1 uses the next hop 2001:db8:acad::1.
- Traffic from R2 to the 2001:db8:acad:209::1 address on R1 uses the next hop 2001:db8:acad:16::1.
- When the G0/0/0 interface on R2 is shut down, traffic from R2 to the 2001:db8:acad:171::1 address on R1 uses the next hop 2001:db8:acad:16::1.

Part 2: Gather Information and Implement Corrections

a. For each criterion that is not met, gather information by examining the running configuration and routing tables to determine what is causing the malfunction.

b. Perform the necessary corrections and record the fix actions.

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1. Traffic from R1 to the 172.16.2.1 address on R2 uses the next hop 192.168.0.14.	
Working as Expected? (Y/N)	N R1# %SYS-5-CONFIG_I: Configured from console by console traceroute 172.16.2.1 Type escape sequence to abort. Tracing the route to 172.16.2.1 1 * * * 2 * * * 3 * * * 4 * * * 5 * * * 6 R1# R1#ping 172.16.2.1 Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds: ..... Success rate is 0 percent (0/5)
Cause of Error	S* 0.0.0.0/0 [1/0] via 192.168.0.3 The default route's next hop ip address is wrong, it should be 192.168.0.14. This is R2's g0/0/0 ip address
Fix Performed	R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.0.14 R1(config)#no ip route 0.0.0.0 0.0.0.0 192.168.0.3 R1#traceroute 172.16.2.1 Type escape sequence to abort. Tracing the route to 172.16.2.1 1 192.168.0.14 0 msec 0 msec 0 msec R1#
2. Traffic from R1 to the 209.165.200.129 address on R2 uses the next hop 192.168.0.30.	
Working as Expected? (Y/N)	N R1#traceroute 209.165.200.129 Type escape sequence to abort. Tracing the route to 209.165.200.129 1 * * * 2 * * * 3 * * * 4 * * * 5 * R1# R1#ping 209.165.200.129 Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 209.165.200.129, timeout is 2 seconds: ..... Success rate is 0 percent (0/5)
Cause of Error	209.165.200.0/24 is variably subnetted, 3 subnets, 2 masks C 209.165.200.0/25 is directly connected, GigabitEthernet0/2 L 209.165.200.1/32 is directly connected, GigabitEthernet0/2 S 209.165.200.128/25 [1/0] via 192.168.0.3 The error is in the last line. The next hop ip address should be 192.168.0.30 not 192.168.0.3

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Fix Performed	R1(config)#no ip route 209.165.200.128 255.255.255.128 192.168.0.3
	R1(config)#ip route 209.165.200.128 255.255.255.128 192.168.0.30
	
	`R1#ping 209.165.200.129`
	
	`Type escape sequence to abort.`
	`Sending 5, 100-byte ICMP Echos to 209.165.200.129, timeout is 2 seconds:`
	`!!!!!`
	`Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms`
	
	`R1#traceroute 209.165.200.129`
	`Type escape sequence to abort.`
	`Tracing the route to 209.165.200.129`
	
	`  1  192.168.0.30  0 msec  0 msec  0 msec`
	`R1#`

3. When the G0/0/0 interface on R1 is shut down, traffic from R1 to the 172.16.2.1 address on R2 uses the next hop 192.168.0.30	
Working as Expected? (Y/N)	N
Cause of Error	When the G0/0/0 interface was shutdown there was no more route for 172.16.2.14. The route got dropped, leaving the traffic nowhere to go

Fix Performed	R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.0.30 10
	The 10 is the administrative distance, so if 192.168.0.14 is up again. 192.168.0.30 will serve as the backup route.
	
	`R1#traceroute 172.16.2.1`
	`Type escape sequence to abort.`
	`Tracing the route to 172.16.2.1`
	
	`  1  192.168.0.30  0 msec  0 msec  0 msec`
	`R1#`
	
	`R1#sh ip route`
	`Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP`
	`  D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area`
	`  N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2`
	`  E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP`
	`  i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area`
	`  * - candidate default, U - per-user static route, o - ODR`
	`  P - periodic downloaded static route`
	
	`Gateway of last resort is 192.168.0.30 to network 0.0.0.0`
	
	`  172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks`
	`C  172.16.1.0/24 is directly connected, GigabitEthernet0/1`
	`L  172.16.1.1/32 is directly connected, GigabitEthernet0/1`
	`  192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks`
	`C  192.168.0.16/28 is directly connected, GigabitEthernet0/0`
	`L  192.168.0.17/32 is directly connected, GigabitEthernet0/0`
	`  209.165.200.0/24 is variably subnetted, 3 subnets, 2 masks`
	`C  209.165.200.0/25 is directly connected, GigabitEthernet0/2`
	`L  209.165.200.1/32 is directly connected, GigabitEthernet0/2`
	`S  209.165.200.128/25 [1/0] via 192.168.0.30`
	`S*  0.0.0.0/0 [10/0] via 192.168.0.30`

4. Traffic from R2 to the 2001:db8:acad:171::1 address on R1 uses the next hop 2001:db8:acad::1.	
Working as Expected? (Y/N)	Y

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Cause of Error	No error
	`R2#ping 2001:db8:acad:171::1`
	
	`Type escape sequence to abort.`
	`Sending 5, 100-byte ICMP Echos to 2001:db8:acad:171::1, timeout is 2 seconds`
	`!!!!!`
	`Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms`
	
	`R2#traceroute 2001:db8:acad:171::1`
	`Type escape sequence to abort.`
	`Tracing the route to 2001:db8:acad:171::1`
	
	`  1  2001:DB8:ACAD::1 1 msec  0 msec  0 msec`
	`R2#`

Fix Performed	N/A
5. Traffic from R2 to the 2001:db8:acad:209::1 address on R1 uses the next hop 2001:db8:acad:16::1.	
Working as Expected? (Y/N)	Y

Cause of Error	No error (like the previous question, it was already working when i tried)
	`R2#ping 2001:db8:acad:209::1`
	
	`Type escape sequence to abort.`
	`Sending 5, 100-byte ICMP Echos to 2001:db8:acad:209::1, timeout is 2 secor`
	`!!!!!`
	`Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms`
	
	`R2#traceroute 2001:db8:acad:209::1`
	`Type escape sequence to abort.`
	`Tracing the route to 2001:db8:acad:209::1`
	
	`  1  2001:DB8:ACAD::1 1 msec  0 msec  0 msec`
	`R2#`

Fix Performed	N/A
6. When the G0/0/0 interface on R2 is shut down, traffic from R2 to the 2001:db8:acad:171::1 address on R1 uses the next hop 2001:db8:acad:16::1.	
Working as Expected? (Y/N)	N

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Cause of Error	Upon checking R2's routing table with "show ipv6 route", R2 does not have a route to 2001:db8:acad:171::/64 network when G0/0/0 is down, so that caused traffic to be forwarded towards an incorrect next-hop (2001:db8:acad:16::1) instead of being dropped.
	`R2#show ipv6 route`
	`IPv6 Routing Table - 7 entries`
	`Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP`
	`U - Per-user Static route, M - MIPv6`
	`I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary`
	`ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect`
	`O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2`
	`ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2`
	`D - EIGRP, EX - EIGRP external`
	`C 2001:DB8:ACAD:16::/64 [0/0]`
	`via GigabitEthernet0/0, directly connected`
	`L 2001:DB8:ACAD:16::2/128 [0/0]`
	`via GigabitEthernet0/0, receive`
	`C 2001:DB8:ACAD:172::/64 [0/0]`
	`via GigabitEthernet0/1, directly connected`
	`L 2001:DB8:ACAD:172::1/128 [0/0]`
	`via GigabitEthernet0/1, receive`
	`C 2001:DB8:ACAD:210::/64 [0/0]`
	`via GigabitEthernet0/2, directly connected`
	`L 2001:DB8:ACAD:210::1/128 [0/0]`
	`via GigabitEthernet0/2, receive`
	`L FF00::/8 [0/0]`
	`via Null0, receive`

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	R2(config)#ipv6 route ::/0 2001:DB8:ACAD:16::1 120 R2(config)#no ipv6 route ::/0 2001:DB8:ACAD:16::1 120 R2(config)#ipv6 route 2001:db8:acad:171::/64 2001:db8:acad::1 R2(config)#ipv6 route 2001:db8:acad:171::/64 2001:db8:acad:16::1 5 R2(config)#exit To ensure proper failover behavior, a floating static IPv6 route was configured on R2 for 2001:db8:acad:171::/64 by setting a primary route via 2001:db8:acad::1 (AD 1) and a backup route via 2001:db8:acad:16::1 with a higher administrative distance of 5. When the G0/0/0 interface is shut down, the primary route is removed and traffic automatically fails over to the floating route. I did show ipv6 route to confirm correct failover and restoration behavior
Fix Performed	Below is the routing table when g0/0/0 is shutdown

```
R2(config)#no ipv6 route ::/0 2001:DB8:ACAD:16::1 120
R2(config)#ipv6 route 2001:db8:acad:171::/64 2001:db8:acad::1
R2(config)#ipv6 route 2001:db8:acad:171::/64 2001:db8:acad:16::1 5
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ipv6 route
IPv6 Routing Table - 11 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S   ::/0 [90/0]
     via 2001:DB8:ACAD::1
C   2001:DB8:ACAD::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD::14/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:DB8:ACAD:16::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:ACAD:16::2/128 [0/0]
     via GigabitEthernet0/0, receive
S   2001:DB8:ACAD:171::/64 [1/0]
     via 2001:DB8:ACAD::1
C   2001:DB8:ACAD:172::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:ACAD:172::1/128 [0/0]
     via GigabitEthernet0/1, receive
C   2001:DB8:ACAD:210::/64 [0/0]
     via GigabitEthernet0/2, directly connected
L   2001:DB8:ACAD:210::1/128 [0/0]
     via GigabitEthernet0/2, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

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```
R2(config)#int g0/0/0
R2(config-if)#shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to down

R2(config-if)#exit
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ipv6 route 2001:db8:acad:171::1
Routing entry for 2001:DB8:ACAD:171::/64
  Known via "static", distance 5, metric 0
  Route count is 1/1, share count 0
  Routing paths:
  Routing paths:2001:DB8:ACAD:16::1
      Last update 2001:DB8:ACAD:16::1 on , 00:00:00 ago
R2#show ipv6 route
IPv6 Routing Table - 8 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C   2001:DB8:ACAD:16::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:ACAD:16::2/128 [0/0]
     via GigabitEthernet0/0, receive
S   2001:DB8:ACAD:171::/64 [5/0]
     via 2001:DB8:ACAD:16::1
C   2001:DB8:ACAD:172::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:ACAD:172::1/128 [0/0]
```

Routing table when g0/0/0 is not shutdown

```
R2(config)#int g0/0/0
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

R2(config-if)#exit
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console
show ipv6 route
IPv6 Routing Table - 11 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S   ::/0 [90/0]
     via 2001:DB8:ACAD::1
C   2001:DB8:ACAD::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD::14/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:DB8:ACAD:16::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:ACAD:16::2/128 [0/0]
     via GigabitEthernet0/0, receive
S   2001:DB8:ACAD:171::/64 [1/0]
     via 2001:DB8:ACAD::1
C   2001:DB8:ACAD:172::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:ACAD:172::1/128 [0/0]
     via GigabitEthernet0/1, receive
C   2001:DB8:ACAD:210::/64 [0/0]
     via GigabitEthernet0/2, directly connected
```

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